```
elif selectresponses=='new':
  print('file id of a Google Responses is part of URL and is obtained from URL as below')
  print('if URL is https://docs.google.com/spreadsheets/d/1WDmI18_XPp0JVW5CmBcimbgf3iY1IggfQ31siqWIp0E/edit#gid=0')
  print('https://docs.google.com/spreadsheets/d/<--------------file id----------------------------->/edit#gid=0')
  print('enter file id as 1WDmI18_XPp0JVW5CmBcimbgf3iY1IggfQ31siqWIp0E ')
  fileid=input('provide file id from URL=')
  csvid=input('provide name of responses file=')
  fileDownloaded = drive.CreateFile({'id': fileid})
  fileDownloaded.GetContentFile(csvid + '.csv', mimetype='text/csv')
  import pandas as pd
  df = pd.read_csv(csvid + '.csv')
  print('Google Responses saved in Pandas dataframe')
else:
  print('error: Enter test or new')

do you want to test the code or work on your new google responses? Answer test or new....new
file id of a Google Responses is part of URL and is obtained from URL as below
if URL is https://docs.google.com/spreadsheets/d/1WDmI18_XPp0JVW5CmBcimbgf3iY1IggfQ31siqWIp0E/edit#gid=0
https://docs.google.com/spreadsheets/d/<--------------file id----------------------------->/edit#gid=0
enter file id as 1WDmI18_XPp0JVW5CmBcimbgf3iY1IggfQ31siqWIp0E
provide file id from URL=1FSRBf8YFZSKDvK5_peSTxk2HCfJlXKkjDAdyjXPwTsQ
provide name of responses file=Copy of OnlineResponses
Google Responses saved in Pandas dataframe
```

Figure 9: First step in customising (Google sheet)

Load Blank Certificate and fonts from Google Drive

```
[10] fileDownloaded = drive.CreateFile({'id':'1UHtxwV3e_OyGQGopLz-PBVt9U1X3quOu'})
     fileDownloaded.GetContentFile('BlankCertificate.jpg')
     fileDownloaded = drive.CreateFile({'id':'1JfMjm5ZffEo2shtZko2LC793GB654iys'})
     fileDownloaded.GetContentFile('font_playfair.ttf')
```

Figure 10: Second step in customising (certificate and font)

Pixel positions can be obtained using any image editor. Ex: MS Paint

```
from PIL import Image, ImageFont, ImageDraw
df_name=df['Name']
df_usn=df['USN']
index = df.index
number_of_rows = len(index)
for i in range(number_of_rows):
  name_text = df_name[i]
  usn_text=df_usn[i]
  my_image = Image.open("BlankCertificate.jpg")
  image_editable = ImageDraw.Draw(my_image)
  name_font = ImageFont.truetype('font_playfair.ttf', 150)
  image_editable.text((250,250), name_text, (128,0,128), font=name_font)
  usn_font = ImageFont.truetype('font_playfair.ttf', 70)
  image_editable.text((250,500), usn_text, (128,0,128), font=usn_font)
```

Figure 11: Third step in customising (pixel positions)

Step 4: The code continues its execution and asks for one last input as in Figure 6. For testing this project, type 'test' (only four small-case letters) and press *Enter*. (After learning the project, customisation of the code can be done by typing 'new'. Customisation of the project is very simple and is discussed in a later section.)

Step 5: The code continues its execution and successfully generates zip file *Certificates_EFY.zip* as shown in Figure 7 and Figure 8. The certificates can be viewed by unzipping the folder using software like WinRAR, or WinZip.

Customisation of the project

For creating your own project, minor changes are to be made in the environment setup depicted in Figure 2. The Python code contains many headings and comments so that beginners in Python can also enjoy this project. Follow the steps given below.

Step 1: Make a copy of the code by using menu bar.

File→Save a Copy in Drive

Step 2: Open your Google drive by visiting the URL *drive.google.com*.

Step 3: Upload your Excel sheet or access Google sheet from Google drive. The sheet should contain information of participants. Click on *share* (View Only) and copy the shared link. Make any changes in the sheet as per your requirement.

Step 4: Run the copied Colab code and select 'new' when prompted. Paste the shared link of Excel/Google sheet and its file name in the fields as shown in Figure 9. In our example, the Excel sheet is named as 'Copy of OnlineResponses' and the shared link is: *https://docs.google.com/spreadsheets/d/1FSRBf8YFZSKDvK5_peSTxk2HCfJlXKkjDAdyjXPwTsQ/edit#gid=0. Here the id is 1FSRBf8YFZSKDvK5_peSTxk2HCfJlXKkjDAdyjXPwTsQ*

Step 5: For changing certificate image and fonts, change the code at various parts shown in Figure 10. Many free fonts and certificates are available online. You can modify the BlankCertificate.jpg image and text fonts. The 'id' parts can be obtained from the URL after sharing the font file and image file.

Step 6: Identify the pixel positions on the certificate where participants' information is to be printed. Change code at highlighted parts shown in Figure 11.

Step 7: Run the code to generate the customised certificates. You should be able to download the certificate in a zipped folder.

Results and future scope

This project is a complete online project and hence can be used with any configuration system, irrespective of operating system and hardware. The project is completely scalable and can be used to generate any number of certificates, any number of times. For intermediate to expert Python coders, the project offers a steep learning curve in the form of Python packages and their utilisation. Simple modifications to code can generate data analysis reports. Interested hobbyists can elaborate the code to create a website for this project using Python code deployment methods. END

By: Vijaykumar Sajjanar and Rajinderkumar M. Math

Both authors are assistant professors, Department of Electronics & Communication, BLDEA's CET, Vijayapur, Karnataka

The article was originally published in the December 2021 issue of Electronics For You.